

Patient-Reported Outcomes

Exploring Recall Periods for EQ-5D-5L and EQ-HWB-9: A Hong Kong Population Study

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ABSTRACT

Objectives: This study aimed to explore the impact of recall period on the measurement properties of the EuroQol 5-Dimension 5-Level (EQ-5D-5L) and EuroQol Health and Well-Being 9 items (EQ-HWB-9) instruments.

Methods: An online cross-sectional survey was conducted with a general population sample in Hong Kong ($n = 1262$). Respondents completed both a “today (TD)” and a “last week (LW)” version of both the EQ-5D-5L and EQ-HWB-9. Psychometric properties (eg, distributional characteristics, ceiling, informativity, convergent validity with conceptually related items, or domains of ReQoL-10 and Patient-Reported Outcomes Measurement Information System – 29 + 2, known-group and structural validity) were assessed for both recall period versions of both instruments.

Results: Considering all measurement properties, differences between LW and TD versions were small but slightly larger for the EQ-HWB-9 than for the EQ-5D-5L. LW versions showed lower ceilings, captured more problems, and demonstrated greater informativity. They had comparable or stronger correlations with ReQoL-10 and Patient-Reported Outcomes Measurement Information System – 29 + 2 items or domains than TD, especially for EQ-HWB-9. Known-group validity was slightly better for LW versions, particularly in mental health subgroups, whereas structural validity of both measures was largely consistent across recall periods.

Conclusions: In this general population sample, although the LW recall versions exhibited slightly superior measurement properties compared with the TD versions, our findings overall support the original recall periods for the EQ-5D-5L and EQ-HWB-9. Future research should evaluate the impacts of varying recall periods across populations and cultural contexts to further refine guidelines for instrument selection in the intended areas of use for each instrument.

Keywords: EQ-5D-5L, EQ-HWB-9, psychometrics, recall period.

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Highlights

- The EQ-5D-5L and EQ-HWB-9 differ in purpose, item content, response scales, and recall periods, with EQ-5D-5L using a ‘today’ recall and EQ-HWB-9 using a ‘last week.’ The contribution of recall periods to observed differences remains unclear.
- This first general population study explores recall period impacts on EQ-5D-5L and EQ-HWB-9 outcomes. Recall period exerts greater impact on EQ-HWB-9 than EQ-5D-5L, with modest effects. ‘Last week’ versions demonstrate slightly superior performance, better capturing temporal variations in health and wellbeing.
- Findings support original recall periods. Further research in intended applications (eg, patients, informal carers, or social care users) across countries and languages is needed to evaluate alternative versions and inform decisions on deviations.

Introduction

Generic health and well-being instruments are widely used to assess patient, carer, care recipient, and population health status and in economic evaluations to guide resource allocation decisions.¹ The most common recall periods in these measures include: “today” or “at present” or “current situation” (eg, EQ-5D, 15D, and ICE CAP), “last week” (eg, EQ Health and Well-being 9 [EQ-HWB-9]), “last 2 weeks” (WEMWBS), “last 4 weeks” (eg, SF-6D), and “unspecified” (eg, AqoL).² There is no single recall period that is optimal for all measures and contexts, including different populations and interventions.³ An appropriate recall period is an important aspect of content validity, ensuring the relevance of the selected items for the target population.⁴

The selection of an appropriate recall period for health and well-being measures is important for ensuring their conceptual

integrity and validity.⁵ Rooted in cognitive psychology, memory decay, in which recollections diminish rapidly over time, may lead to underreporting of symptoms or experiences in longer periods, such as “last week” (LW) or “last 4 weeks,” potentially undermining accuracy for fluctuating constructs, such as pain or emotional well-being.^{3,6} Conversely, the averaging of experiences over extended intervals allows respondents to synthesize highs and lows into a composite appraisal, offering a broader view but risking dilution of day-to-day variations or overemphasis on peaks.⁷ Priming effects may further complicate measurement accuracy because recent or salient events can disproportionately shape recall, biasing reports toward current moods or memorable incidents and altering perceived status. Additional biases, such as telescoping, may also exacerbate distortions in longer recalls, highlighting the need to align periods with the phenomenon’s characteristics to mitigate these mechanisms and enhance content validity.⁸

Table 1. Sample characteristics.

Characteristics	n (%)	HK general population (%)	χ^2 test
Age groups (years)			$P = 1.00$
20-24	78 (6.2)	6.1	
25-29	114 (9.0)	8.5	
30-34	129 (10.2)	9.8	
35-39	155 (12.3)	10.9	
40-44	143 (11.3)	10.9	
45-49	127 (10.1)	10.8	
50-54	130 (10.3)	10.7	
55-59	138 (11.0)	11.7	
60-64	135 (10.7)	11.4	
65-69	113 (9.0)	9.2	
Gender			$P = 1.00$
Male	567 (44.9)	45.6	
Female	695 (55.1)	54.4	
Marital status			
Married	740 (58.6)		
Never married	458 (36.3)		
Divorced/widowed	64 (5.1)		
Educational level			
Secondary and below	514 (40.7)		
Postsecondary and above	748 (59.3)		
Perceived income level			
Below local average	234 (18.5)		
Local average	764 (60.5)		
Above local average	264 (20.9)		
Employment			
Full-time	933 (73.9)		
Part-time	113 (9.0)		
Others	216 (17.1)		
Living condition			
Alone	146 (11.6)		
With others	1116 (88.4)		
Informal caregiver			
Yes	323 (25.6)		
No	939 (74.4)		
Chronic diseases			
Yes	324 (25.7)		
No	829 (65.7)		
Don't want to answer the question	109 (8.6)		

HK general population data, <https://www.census2021.gov.hk/en/index.html>.

EuroQol instruments, particularly the EQ-5D, are the most widely used among generic preference-accompanied measures and are also recommended in health technology assessment

guidelines in a large number of countries.^{9,10} Among EuroQol instruments targeting adults, the EQ-5D instruments (EQ-5D-3L and EQ-5D-5L) use a “today” recall period and focus on the respondent’s health on the day of completion.¹¹ The currently experimental instrument, EQ-HWB-9, was developed to assess health, social care, and carer-related quality of life and uses a “last week” recall period.¹² Both the EQ-5D-5L and EQ-HWB-9 descriptive systems share some similar item content; for example, both include physical functioning, daily activities, pain, and mental health items. However, the EQ-HWB-9 also captures broader aspects, such as loneliness, exhaustion, cognition, and control. This distinction indicates the importance of testing recall periods to assess whether the EQ-HWB-9’s LW recall period more effectively captures cumulative health and well-being experiences. Furthermore, the extent to which differences in recall periods contribute to variations in findings between the 2 instruments remains unknown. For the EQ-5D-5L, existing research on alternative recall periods has predominantly focused on patient populations, often with relatively small sample sizes.^{13,14} For the EQ-HWB-9, there is an absence of studies investigating different recall periods together with the EQ-5D-5L.^{15,16} Thus, this study aimed to explore the impact of different recall periods on the measurement properties of the EQ-5D-5L and EQ-HWB-9.

Methods

Data Collection and Participants

This study used data from a web-based cross-sectional survey, conducted among members of Hong Kong general population between January and March 2025. Participants were recruited through the online panel of a local survey company. Inclusion criteria were (1) being aged between 20 and 69 years, (2) being able to read and speak Chinese, and (3) being able to provide informed consent. The exclusion criteria were as follows: (1) failure to complete the Completely Automated Public Turing test to tell Computers and Humans Apart questions, (2) being identified as bots by the Qualtrics platform, (3) completing the questionnaire in less than 6 minutes or more than 30 minutes, (4) having duplicate IP addresses, and (5) failure to provide consistent responses to cross-validated questions. We aimed to obtain a representative sample using quota sampling based on sex and age. The study protocol was approved by the ethics committee of the Institutional Review Board of the Hong Kong Polytechnic University (ref No.: HSEARS20240917004), and written informed consent was obtained from all participants.

Outcome Measures

EuroQol instruments

All EuroQol instruments and their modifications were registered, and a permission-to-modify agreement was signed with the EuroQol Office. This study included the EQ-5D-5L¹¹ and the experimental version of EQ-HWB-9 (2022 version, EV1.0).¹² The standard EQ-5D-5L descriptive system uses a “today” (TD) recall period, whereas the EQ-HWB-9 uses a LW recall period. Alongside the standard versions, 2 modifications were developed: a “LW” version of the EQ-5D-5L and a “TD” version of the EQ-HWB-9. For both EuroQol instruments, we calculated level sum scores (LSSs), which were transformed to a 0 to 100 scale with higher scores indicating better outcomes. Results for EQ VAS are not covered in the present article and will be reported elsewhere.

Table 2. Distribution of responses on and measurement profile of the EQ-5D-5L and EQ-HWB-9.

Instruments	n (%)											P value*
	Today					Last week						
	1	2	3	4	5	1	2	3	4	5		
EQ-5D-5L												
Mobility	1079 (85.5)	141 (11.2)	36 (2.9)	6 (0.5)	0 (0.0)	1066 (84.5)	156 (12.4)	31 (2.5)	5 (0.4)	4 (0.3)	.282	
Self-care	1154 (91.4)	79 (6.3)	25 (2.0)	4 (0.3)	0 (0.0)	1148 (91.0)	91 (7.2)	17 (1.3)	6 (0.5)	0 (0.0)	.423	
Usual activities	1094 (86.7)	125 (9.9)	31 (2.5)	9 (0.7)	3 (0.2)	1096 (86.8)	124 (9.8)	36 (2.9)	4 (0.3)	2 (0.2)	.651	
Pain/discomfort	618 (49.0)	545 (43.2)	83 (6.6)	14 (1.1)	2 (0.2)	529 (41.9)	620 (49.1)	95 (7.5)	17 (1.3)	1 (0.1)	.007	
Anxiety/depression	673 (53.3)	438 (34.7)	103 (8.2)	30 (2.4)	18 (1.4)	634 (50.2)	472 (37.4)	112 (8.9)	28 (2.2)	16 (1.3)	.560	
11,111	439 (34.8)	-	-	-	-	364 (28.8)	-	-	-	-	.002	
Agreement of profiles between TD and LW												
TD = LW	661 (52.4)	-	-	-	-	-	-	-	-	-		
TD = LW (excluding 11,111)	353 (28.0)	-	-	-	-	-	-	-	-	-		
TD better than LW	177 (14.0)	-	-	-	-	-	-	-	-	-		
TD worse than LW	321 (25.4)	-	-	-	-	-	-	-	-	-		
Mixed	103 (8.2)	-	-	-	-	-	-	-	-	-		
EQ-HWB-9												
Getting around inside and outside	1090 (86.4)	123 (9.7)	34 (2.7)	14 (1.1)	1 (0.1)	1056 (83.7)	154 (12.2)	38 (3.0)	11 (0.9)	3 (0.2)	.230	
Day-to-day activities	1047 (83.0)	149 (11.8)	47 (3.7)	15 (1.2)	4 (0.3)	1011 (80.1)	192 (15.2)	46 (3.6)	12 (1.0)	1 (0.1)	.082	
Exhausted	466 (36.9)	531 (42.1)	191 (15.1)	57 (4.5)	17 (1.3)	405 (32.1)	563 (44.6)	208 (16.5)	75 (5.9)	11 (0.9)	.043	
Lonely	610 (48.3)	406 (32.2)	167 (13.2)	56 (4.4)	23 (1.8)	518 (41.0)	476 (37.7)	194 (15.4)	55 (4.4)	19 (1.5)	.004	
Concentrating/ thinking clearly	593 (47.0)	433 (34.3)	180 (14.3)	42 (3.3)	14 (1.1)	515 (40.8)	493 (39.1)	198 (15.7)	47 (3.7)	9 (0.7)	.022	
Anxious	579 (45.9)	421 (33.4)	171 (13.5)	65 (5.2)	26 (2.1)	488 (38.7)	459 (36.4)	208 (16.5)	82 (6.5)	25 (2.0)	.005	
Depressed	646 (51.2)	371 (29.4)	173 (13.7)	49 (3.9)	23 (1.8)	563 (44.6)	420 (33.3)	203 (16.1)	49 (3.9)	27 (2.1)	.021	
Control	628 (49.8)	383 (30.3)	171 (13.5)	62 (4.9)	18 (1.4)	578 (45.8)	419 (33.2)	178 (14.1)	67 (5.3)	20 (1.6)	.399	
Pain (severity)	607 (48.1)	561 (44.5)	78 (6.2)	15 (1.2)	1 (0.1)	481 (38.1)	656 (52.0)	112 (8.9)	10 (0.8)	3 (0.2)	<.001	
111,111,111	230 (18.2)	-	-	-	-	184 (14.6)	-	-	-	-	.024	
Agreement of profiles between TD and LW												
TD = LW	223 (17.7)	-	-	-	-	-	-	-	-	-		
TD = LW (excluding 111111111)	89 (7.1)	-	-	-	-	-	-	-	-	-		
TD better than LW	226 (17.9)	-	-	-	-	-	-	-	-	-		
TD worse than LW	426 (33.8)	-	-	-	-	-	-	-	-	-		
Mixed	387 (30.7)	-	-	-	-	-	-	-	-	-		
Measurement [†]			Mean (SD)	Median (IQR)	Range	Ceiling (%)	Difference of mean	P value [†]				
EQ-5D-5L LSS (TD)			13.20 (16.84)	7.69 (15.38)	(0.00, 100.00)	34.8	−2.10	.020				
EQ-5D-5L LSS (LW)			15.30 (17.28)	8.33 (16.67)	(0.00, 100.00)	28.8						
EQ-HWB-9 LSS (TD)			19.64 (18.75)	16.67 (26.67)	(0.00, 100.00)	18.2	−3.84	<.001				
EQ-HWB-9 LSS (LW)			23.48 (19.81)	21.43 (25.00)	(0.00, 100.00)	14.6						

continued on next page

Table 2. Continued

Measurement [‡]	Mean (SD)	Median (IQR)	Range	Ceiling (%)	Difference of mean	P value [†]
EQ-5D-5L LSS (TD) vs EQ-HWB-9 LSS (TD)	-	-	-	-	-6.44	<.001
EQ-5D-5L LSS (TD) vs EQ-HWB-9 LSS (LW)	-	-	-	-	-10.29	<.001
EQ-5D-5L LSS (LW) vs EQ-HWB-9 LSS (TD)	-	-	-	-	-4.34	<.001
EQ-5D-5L LSS (LW) vs EQ-HWB-9 LSS (LW)	-	-	-	-	-8.18	<.001

LSS indicates level sum scores; LW, last week; TD, today; P value for all correlation coefficients is <.001.

*Chi-Square test, Yates' correction was applied when any cell value was less than 5.

[†]P value was calculated based on repeated measures ANOVA, followed by Tukey's honestly significant difference post hoc test.

[‡]Only comparisons of mean values between measures that showed significant differences are presented.

Patient-reported outcomes measurement information system ([PROMIS]) – 29 + 2

PROMIS – 29 + 2 measures health across 8 domains.^{17,18} It uses a 7-day recall period for most of its domains, except for physical function and ability to participate in social roles and activities, which do not specify a recall period.

Recovering quality of life (ReQoL)-10

The ReQoL-10 assesses quality of life in individuals with mental health conditions using a 1-week recall period and a 5-point scale, with higher scores reflecting better quality of life.¹⁹

Design of Questionnaire

The questionnaire was programmed in Qualtrics (Provo, UT, 2020). The first page presented the informed consent form, which participants had to read and agree to before proceeding with the survey. The questionnaire was structured into 4 sections. In section 1, respondents completed 2 of the 4 variants in a randomized order: either an EQ-5D-5L variant ("TD" or "LW") followed by an EQ-HWB-9 variant ("TD" or "LW"), or the reverse. In section 2, they completed PROMIS – 29 + 2 and ReQoL-10 in a randomized sequence. Section 3 then presented the 2 remaining variants of EuroQol instruments not included in section 1 (eg, if "TD" EQ-5D-5L and "LW" EQ-HWB-9 appeared in section 1, then "LW" EQ-5D-5L and "TD" EQ-HWB-9 followed), also in a randomized order. Section 4 collected background information. This design yielded 16 possible randomization arms to minimize ordering bias. The headlines of 4 variants of the EQ-HWB-9 and EQ-5D, in both English and Chinese, are presented in [Appendix 1](#) in [Supplemental Materials](#). To draw respondents' attention, the recall period was highlighted using different colors and bold text. The modified versions are available for further methodological research and can be obtained directly from the EuroQol Research Foundation. Although the overall study design was cross-sectional, with all assessments conducted at a single time point, the data set incorporates repeated measures because each participant completed both versions of both instruments (a total of 4 4 versions).

Data Analysis

Data analysis was performed using R software (version 4.5.1). Before the analysis, data preprocessing was conducted in Microsoft Excel to identify and exclude invalid responses based on pre-defined criteria. Because all survey questions were mandatory, respondents were prevented from advancing without completing prior items, there were no missing values in the data set.

Descriptive characteristics

The representativeness of the sample was checked against the Hong Kong general population for sex and age using a chi-square test. The relative frequency of responses for each response level of each item in the descriptive systems of the EQ-5D-5L and EQ-HWB-9 was computed for both recall period versions. A chi-square test with Yates' correction was used to compare differences in response distributions between the TD and LW versions of each instrument. Differences in LSSs of EQ-5D-5L and EQ-HWB-9 across the 4 versions were analyzed using repeated measures analysis of variance, followed by Tukey's honestly significant difference post hoc test. Histograms were used to visually represent the distributions of these scores and their differences between TD and LW.

Ceiling and floor

Ceiling and floor were examined at both the item and scale levels to assess the items and the instrument's sensitivity to differences in health and well-being at both ends of the items or scale and to determine how this is affected by the recall period. We hypothesized a lower overall ceiling on the EQ-HWB-9 descriptive system compared with the EQ-5D-5L for both the TD and LW recall periods, given the higher number of items in the EQ-HWB-9 descriptive system.^{15,16,20,21} Based on the literature, we also hypothesized a lower ceiling on the LW versions compared with the TD versions.⁷

Informativity

Informativity quantifies response distribution evenness across levels, revealing if longer recalls lead to a greater variability in profiles compared with immediate recalls. We computed the Shannon index (H') and the Shannon evenness index (J') to assess the absolute and relative informativity of each item of the descriptive system, respectively, for both the TD and LW versions of the EQ-5D-5L and EQ-HWB-9.²²

Profile consistency and agreement

Profile consistency across recall periods was evaluated to quantify systematic differences and agreement between recall periods. Respondents were grouped into 4 categories for the EQ-5D-5L and EQ-HWB-9 descriptive systems: (1) uniform (identical TD and LW profiles), (2) uniform excluding 11,111 (EQ-5D-5L) or 1111111 (EQ-HWB-9), (3) worse (LW equal or worse in all items), (4) better (LW equal or better in all items), and (5) mixed (LW better in at least one item, worse in another). Bland-Altman plot and intraclass correlation coefficients were used to

Table 3. Convergent validity of EQ-5D-5L and EQ-HWB-9.

Instruments	Instruments	Correlation coefficients	
Item-level associations		TD	LW
EQ-5D-5L mobility	EQ-HWB-9 getting around inside and outside (TD)	0.60	0.55
	EQ-HWB-9 getting around inside and outside (LW)	0.60	0.58
	PROMIS – 29 + 2 Physical function*	0.56	0.55
EQ-5D-5L Usual activities	EQ-HWB-9 Day-to-day activities (TD)	0.55	0.59
	EQ-HWB-9 Day-to-day activities (LW)	0.55	0.55
	PROMIS – 29 + 2 Physical function*	0.56	0.56
	ReQoL-10 Everyday tasks	0.22	0.24
EQ-5D-5L pain/discomfort	EQ-HWB-9 Pain (severity) (TD)	0.73	0.70
	EQ-HWB-9 Pain (severity) (LW)	0.71	0.74
	PROMIS – 29 + 2 Pain intensity NRS*	0.60	0.62
	PROMIS – 29 + 2 Pain interference*	0.54	0.54
EQ-5D-5L anxiety/depression	EQ-HWB-9 anxiety (TD)	0.63	0.65
	EQ-HWB-9 anxiety (LW)	0.66	0.68
	EQ-HWB-9 sadness/depressed (TD)	0.61	0.61
	EQ-HWB-9 sadness/depressed (LW)	0.64	0.67
	PROMIS – 29 + 2 anxiety*	0.68	0.70
	PROMIS – 29 + 2 depression*	0.67	0.70
	ReQoL-10 I felt happy	0.37	0.38
	ReQoL-10 I felt confident in myself	0.35	0.34
	ReQoL-10 Life was not worth living	0.45	0.48
EQ-HWB-9 getting around inside and outside	PROMIS – 29 + 2 Physical function*	0.54	0.59
EQ-HWB-9 Day-to-day activities	PROMIS – 29 + 2 Physical function*	0.56	0.61
EQ-HWB-9 exhausted	PROMIS – 29 + 2 fatigue*	0.67	0.68
EQ-HWB-9 lonely	ReQoL-10 I felt lonely	0.63	0.70
EQ-HWB-9 Concentrating or thinking clearly	PROMIS – 29 + 2 cognition*	0.23	0.22
EQ-HWB-9 anxiety	PROMIS – 29 + 2 anxiety*	0.69	0.75
EQ-HWB-9 sadness/depressed	PROMIS – 29 + 2 depression*	0.68	0.76
	ReQoL-10 I felt happy	0.35	0.38
	ReQoL-10 I thought my life was not worth living	0.48	0.54
EQ-HWB-9 control	ReQoL-10 I felt unable to cope	0.52	0.56
EQ-HWB-9 Pain (severity)	PROMIS – 29 + 2 Pain intensity NRS*	0.64	0.67
	PROMIS – 29 + 2 Pain interference*	0.59	0.58
Scale-level associations			
EQ-5D-5L LSS (TD)	ReQoL LSS	0.41	
EQ-5D-5L LSS (LW)	ReQoL LSS	0.42	
EQ-HWB-9 LSS (TD)	ReQoL LSS	0.62	
EQ-HWB-9 LSS (LW)	ReQoL LSS	0.65	
EQ-5D-5L LSS (TD)	PROMIS – 29 + 2LSS	0.57	
EQ-5D-5L LSS (LW)	PROMIS – 29 + 2LSS	0.59	
EQ-HWB-9 LSS (TD)	PROMIS – 29 + 2LSS	0.74	
EQ-HWB-9 LSS (LW)	PROMIS – 29 + 2LSS	0.79	
EQ-5D-5L LSS (TD)	EQ-HWB LSS (TD)	0.67	
EQ-5D-5L LSS (TD)	EQ-HWB LSS (LW)	0.69	
EQ-5D-5L LSS (LW)	EQ-HWB LSS (TD)	0.70	
EQ-5D-5L LSS (LW)	EQ-HWB LSS (LW)	0.70	
EQ-5D-5L LSS (TD)_	EQ-5D-5L LSS (LW)_	0.82	
EQ-HWB LSS (TD)	EQ-HWB LSS (LW)	0.81	

EQ-HWB indicates EQ Health and Well-Being; LSS, level sum scores; LW, Last week; NRS, numeric rating scale; PROMIS, Patient-Reported Outcomes Measurement Information System; TD, today; *P* value for all correlation coefficients is < .001.

*Raw domain score of PROMIS – 29 + 2 was used; correlation strength: > 0.5, strong; 0.31 to 0.5, moderate; ≤ 0.3, weak.

Table 4. Known-group validity of EQ-5D-5L and EQ-HWB-9.

Indicators	n	EQ-5D-5L LSS		EQ-HWB-9 LSS	
		TD	LW	TD	LW
General health status					
Excellent	112	7.83 (17.91)	8.56 (17.13)	12.62 (19.14)	12.63 (17.01)
Very good	454	8.07 (13.19)	9.54 (12.93)	12.23 (13.14)	15.54 (14.81)
Good	382	13.03 (15.83)	14.86 (14.70)	19.04 (16.54)	24.00 (17.55)
Fair	278	20.78 (16.91)	24.79 (19.32)	31.24 (18.80)	35.43 (19.31)
Poor	36	37.82 (21.00)	40.28 (20.65)	51.67 (22.40)	59.72 (19.78)
F-statistics		54.72	68.88	97.33	110.80
P value		<.001	<.001	<.001	<.001
η ²		0.15	0.18	0.24	0.26
Informal caregiver					
Yes	323	17.03 (18.92)	18.11 (17.27)	24.14 (20.38)	29.11 (20.41)
No	939	11.88 (15.85)	14.33 (17.19)	18.09 (17.91)	21.55 (19.23)
t statistics		4.78	3.4	5.05	6
P value		<.001	<.001	<.001	<.001
Cohen's d		0.31	0.22	0.33	0.39
Perceived income level					
Below local average	234	21.43 (21.38)	24.68 (22.19)	31.04 (22.75)	37.74 (22.02)
Local average	764	11.12 (14.32)	13.10 (14.75)	17.78 (16.40)	21.30 (17.56)
Above local average	264	11.92 (16.87)	13.35 (16.39)	14.90 (17.28)	17.15 (17.95)
F-statistics		36.55	45.31	60.66	89.85
P value		<.001	<.001	<.001	<.001
η ²		0.05	0.07	0.09	0.15
Sleep quality					
Excellent	141	7.53 (14.43)	7.68 (14.16)	11.30 (15.16)	12.74 (14.94)
Very good	341	8.57 (14.82)	10.41 (14.84)	12.15 (14.12)	15.53 (15.97)
Good	303	11.42 (14.72)	12.84 (13.78)	17.37 (15.77)	21.91 (17.47)
Fair	377	17.16 (17.47)	20.78 (18.49)	25.56 (18.69)	29.42 (18.35)
Poor	100	27.38 (18.69)	29.50 (19.66)	41.47 (21.47)	48.11 (21.46)
F-statistics		38.29	47.71	81.53	92.50
P value		<.001	<.001	<.001	<.001
η ²		0.11	0.13	0.21	0.23
Smoking					
Yes/Former smoker	344	16.08 (18.99)	18.22 (20.53)	22.87 (21.31)	26.54 (21.90)
Never	915	12.12 (15.86)	14.13 (15.70)	18.33 (17.48)	22.26 (18.81)
t statistics		3.73	3.77	3.86	3.43
P value		<.001	<.001	<.001	<.001
Cohen's d		0.24	0.24	0.24	0.22
Alcohol consumption					
≤ 1 per month	725	10.96 (13.85)	13.36 (14.60)	18.47 (17.73)	22.49 (18.94)
≥2 per month	497	16.33 (20.11)	18.09 (20.23)	21.39 (20.17)	24.76 (20.99)
t statistics		−5.53	−4.75	−2.67	−1.97
P value		<.001	<.001	.008	.049
Cohen's d		0.32	0.28	0.16	0.11
Diet					
Very healthy	89	10.20 (17.73)	9.27 (15.41)	9.78 (14.02)	12.88 (16.96)
Healthy	334	9.37 (17.29)	11.18 (16.15)	14.07 (16.52)	16.55 (18.04)

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Table 4. Continued

Indicators	n	EQ-5D-5L LSS		EQ-HWB-9 LSS	
		TD	LW	TD	LW
Fairly healthy	442	11.29 (12.47)	13.56 (14.07)	17.90 (16.04)	21.33 (15.99)
Average	366	18.28 (18.45)	21.13 (19.12)	26.99 (19.89)	32.77 (20.22)
Unhealthy	31	30.02 (19.90)	33.06 (22.82)	45.91 (23.11)	49.65 (24.73)
F-statistics		24.18	29.66	50.59	61.40
P value		<.001	<.001	<.001	<.001
η^2		0.07	0.09	0.14	0.16
Exercise					
≤90 minutes/week	498	14.86 (17.22)	17.47 (18.79)	23.39 (20.25)	27.81 (20.55)
>90 minutes/week	764	12.11 (16.51)	13.89 (16.08)	17.19 (17.29)	20.66 (18.80)
t statistics		2.84	3.62	5.82	6.36
P value		.005	<.001	<.001	<.001
Cohen's d		0.16	0.21	0.34	0.37
Living with chronic conditions					
Yes	324	17.45 (18.41)	20.37 (19.90)	23.96 (20.68)	29.53 (20.64)
No	829	10.71 (15.07)	12.46 (14.75)	16.53 (16.68)	19.93 (18.21)
t statistics		6.40	7.37	6.34	7.74
P value		<.001	<.001	<.001	<.001
Cohen's d		0.42	0.48	0.42	0.51
Hypertension					
Yes	78	17.55 (15.92)	20.09 (16.03)	24.06 (17.20)	30.04 (19.36)
No	829	10.71 (15.07)	12.46 (14.75)	16.53 (16.68)	19.93 (18.21)
t statistics		3.82	4.33	3.80	4.66
P value		<.001	<.001	<.001	<.001
Cohen's d		0.45	0.51	0.45	0.55
Hyperlipidemia					
Yes	65	21.78 (17.23)	24.49 (19.26)	26.72 (18.94)	32.69 (19.64)
No	829	10.71 (15.07)	12.46 (14.75)	16.53 (16.68)	19.93 (18.21)
t statistics		5.64	6.17	4.69	5.41
P value		<.001	<.001	<.001	<.001
Cohen's d		0.73	0.80	0.60	0.70
Visual problems					
Yes	40	25.00 (23.35)	29.79 (25.45)	34.92 (28.33)	36.34 (25.85)
No	829	10.71 (15.07)	12.46 (14.75)	16.53 (16.68)	19.93 (18.21)
t statistics		5.68	6.95	6.54	5.44
P value		<.001	<.001	<.001	<.001
Cohen's d		0.92	1.13	1.06	0.88
Skin diseases (eg, psoriasis, eczema, and urticaria)					
Yes	35	18.24 (15.17)	20.95 (16.59)	27.33 (22.85)	32.76 (22.26)
No	829	10.71 (15.07)	12.46 (14.75)	16.53 (16.68)	19.93 (18.21)
t statistics		2.90	3.32	3.69	4.04
P value		.004	<.001	<.001	<.001
Cohen's d		0.50	0.57	0.64	0.70
Mental health problems					
Yes	40	35.19 (20.31)	36.67 (25.93)	43.25 (23.15)	52.05 (21.91)
No	829	10.71 (15.07)	12.46 (14.75)	16.53 (16.68)	19.93 (18.21)

continued on next page

Table 4. Continued

Indicators	n	EQ-5D-5L LSS		EQ-HWB-9 LSS	
		TD	LW	TD	LW
t statistics		9.85	9.69	9.69	10.79
P value		<.001	<.001	<.001	<.001
Cohen's d		1.60	1.57	1.57	1.75

LSS indicates level sum scores; LW, last week; TD, today.

assess agreement between TD and LW for the EQ-5D-5L and EQ-HWB-9 LSSs.

Convergent validity

Convergent validity of the different recall period versions was examined to determine whether the recall period affects the items' and the instrument's ability to consistently capture the key underlying constructs. Convergent validity was assessed by testing hypotheses (a full list is provided in [Appendix 2](#) in [Supplemental Materials](#)) about correlations between the EQ-5D-5L and EQ-HWB-9 (both recall periods) and corresponding the PROMIS – 29 + 2 and ReQoL-10 items or domains measuring similar constructs. We hypothesized at least moderate correlations. Stronger correlations were expected between LW variants of the EQ-5D-5L/EQ-HWB-9 and PROMIS – 29 + 2/ReQoL – 10, which shared the same recall period, compared with TD variants. Similarly, strong correlations were anticipated between the EQ-5D-5L and EQ-HWB-9 items assessing similar constructs, with higher coefficients for matching recall periods. Spearman's correlation coefficients were computed and categorized as weak (<0.3), moderate (0.3–0.5), or strong (>0.5).²³ Pearson's correlation coefficients assessed relationships between the EQ-5D-5L and EQ-HWB-9 LSSs, PROMIS – 29 + 2 LSS and ReQoL-10 LSS, using identical thresholds. Moderate correlations were expected based on the literature.^{24–26}

Known-group validity

Known-group validity was tested to examine whether the recall period influences the instrument's ability to detect expected differences between groups of respondents. Independent samples t tests (for 2 groups) and 1-way analysis of variance (for more than 2 groups) were used to evaluate whether LSSs were able to distinguish between predefined groups of respondents. Cohen's d (2-group comparison, ≥ 0.8 large, 0.5–0.79 medium, 0.2–0.49 small, and <0.2, trivial) and η^2 (3-group comparison, ≥ 0.14 large, 0.06–0.13 medium, and 0.01–0.05, small) were calculated to reflect the measures' effect size.²⁷ We hypothesized that individuals with poor general health, chronic conditions, informal caregiving responsibilities, lower income, poor sleep quality, or unhealthy lifestyle behaviors (eg, smoking, excessive alcohol consumption, poor diet, or insufficient exercise) would report lower health and well-being. For chronic conditions, each disease group was compared with healthy respondents.

Structural validity

Structural validity assessment was used to empirically understand the underlying structure of the instruments and evaluate their performance consistency across different recall periods. Because this study introduced varying recall periods ("TD" and "LW") for the EQ-5D-5L and EQ-HWB-9, modifications with limited

prior empirical evidence, we selected principal-component analysis (PCA) as an exploratory technique. This approach, used in previous EQ instrument studies,^{28–33} enabled us to investigate the dimensional structure and facilitate direct comparisons with the more reflective PROMIS – 29 + 2 and ReQoL-10. Accordingly, we conducted PCA on the items from 4 instruments (EQ-5D-5L, EQ-HWB-9, PROMIS – 29 + 2, and ReQoL-10) for the TD and LW versions separately, aiming to identify underlying constructs and detect any variations in factor structure attributable to the altered recall periods in the EuroQol instruments. We examined whether factor structures varied across recall periods for EuroQol instruments. Data adequacy was assessed using Kaiser-Meyer-Olkin (KMO > 0.9) and Bartlett's tests ($P < .05$).³⁴ Varimax rotation was applied, retaining factors with eigenvalues > 1.³⁵ The factor loading cutoff was set at ≥ 0.4 .³⁶

Results

Participants' Background Characteristics

In total, 1262 participants' data were analyzed. Of the 2168 individuals who accessed the survey link, 1682 completed it, with 420 excluded for failing to meet data quality criteria. Sample characteristics are presented in [Table 1](#). No significant differences were observed in the background characteristics of respondents across the 16 study arms (see [Appendix 3](#) in [Supplemental Materials](#)).

Descriptive Characteristics, Ceiling, and Floor

[Table 2](#) presents the proportion reporting the best possible response of the EQ-5D-5L was lower for LW (41.9%–91.0%) than for TD (49.0%–91.4%). Similarly, the EQ-HWB-9 exhibited a lower ceiling for LW (32.1%–83.7%) compared with TD (36.9%–86.4%). Differences in response distributions between TD and LW versions were statistically significant only for the pain/discomfort item in the EQ-5D-5L. For the EQ-HWB-9, all items showed significant differences except for "getting around inside and outside" and "control." For EQ-5D-5L, the best possible profile was reported by 34.8% (TD) and 28.8% (LW) ($P = .002$). For EQ-HWB-9, it was 18.2% (TD) and 14.6% (LW) ($P = .024$).

The mean LSSs of the EQ-HWB-9 exceeded those of the EQ-5D-5L LSSs across recall periods. The largest mean LSS difference was between EQ-HWB-9 (LW) and EQ-5D-5L (TD). For EQ-5D-5L, 52.4% showed consistent TD and LW profiles, dropping to 28.0% excluding full health profiles. EQ-HWB-9 consistency was lower, with 17.7% identical profiles and 7.1% excluding best profiles, indicating greater variability ([Table 2](#)). The distributions of LSSs and B-A plot with ICC values are presented in [Appendices 4 and 5](#) in [Supplemental Materials](#).

Table 5. Results of the principal-component analysis.

Instrument	Mental health		Physical health		Pain		Cognition and QoL		Ability to participate in social roles and activities		Sleep quality
	LW	TD	LW	TD	LW	TD	LW	TD	LW	TD	TD
EQ-5D-5L											
Mobility			0.72	0.77							
Self-care			0.71	0.72							
Usual activities			0.72	0.75							
Pain/discomfort					0.66	0.67					
Anxiety/depression	0.75	0.70									
EQ-HWB-9											
Getting around inside and outside			0.77	0.74							
Day-to-day activities			0.71	0.73							
Exhausted	0.64	0.67									
Lonely	0.76	0.76									
Concentrating/thinking clearly	0.66	0.70									
Anxiety	0.81	0.80									
Sadness/depressed	0.81	0.78									
Control	0.75	0.75									
Pain (severity)					0.70	0.69					
PROMIS – 29 + 2											
Do chores such as vacuuming or yard work			0.74	0.71							
Go up and down stairs at a normal pace			0.70	0.66							
Go for a walk of at least 15 min			0.78	0.79							
Run errands and shop			0.78	0.78							
Felt fearful	0.78	0.77									
Found it hard to focus on anything other than my anxiety	0.75	0.75									
Worries overwhelmed one	0.78	0.78									
Felt uneasy	0.80	0.79									
Felt worthless	0.79	0.78									
Felt helpless	0.80	0.79									
Felt depressed	0.84	0.83									
Felt hopeless	0.77	0.76									
Felt fatigued	0.64	0.65									
Have trouble starting things because being tired	0.68	0.69									
Felt run-down on average	0.71	0.72									
Fatigued on average	0.73	0.73									
Sleep quality					0.46						0.57
Refreshing sleep					0.43						0.45
Problem with one's sleep					0.44						0.50
Difficulty falling asleep					0.42						0.49
Trouble doing all of one's regular leisure activities with others									0.58	0.60	
Trouble doing all of the family activities that one want to do									0.65	0.67	
Trouble doing all of one's usual work (include work at home)									0.57	0.60	
Trouble doing all of the activities with friends that one want to do									0.62	0.64	
Pain inter-reference with one's day-to-day activities					0.64	0.70					
Pain interference with work around the home					0.57	0.65					
Pain interference with one's ability to participate in social activities					0.46	0.53					
Pain interference with one's household chores					0.55	0.60					
Concentration							0.58	0.39			
Remembering to do things, such as take medicine or buy something one needed							0.57	0.37			
Average pain 0-10 numeric rating scale					0.67	0.69					

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Table 5. Continued

Instrument	Mental health		Physical health		Pain		Cognition and QoL		Ability to participate in social roles and activities		Sleep quality
	LW	TD	LW	TD	LW	TD	LW	TD	LW	TD	TD
ReQoL-10											
Found it difficult to get start with everyday tasks	0.60	0.60									
Felt unable to trust others							0.65	0.65			
Felt unable to cope											
Found do the things one want to do							0.73	0.75			
Felt happy							0.77	0.80			
Thought my life was not worth living	0.62	0.60									
Enjoyed what I did							0.79	0.82			
Felt hopeful about my future							0.74	0.77			
Felt lonely	0.67	0.66									
Felt confident in my life							0.74	0.77			

LW indicates last week; PROMIS, Patient-Reported Outcomes Measurement Information System; TD, today; *P* value for all correlation coefficients is < .001.

Convergent Validity

Most comparisons of similar constructs of external measures showed at least moderate correlations with their corresponding EQ-5D-5L or EQ-HWB-9 items, regardless of the recall period (Table 3). For the EQ-HWB-9, LW items consistently exhibited higher correlation coefficients than those with the TD. For the EQ-5D-5L, the pattern was mixed: mobility tended to show higher correlation coefficients with the TD recall period

compared with LW, whereas the opposite was true for the anxiety/depression item. The correlation between PROMIS – 29 + 2 cognition domain score and EQ-HWB-9 “concentrating/thinking clearly” was statistically significant but weak. At the instrument level, moderate or strong correlations were observed across all measures (Table 3). The LW versions of EQ-5D-5L and EQ-HWB-9 exhibited slightly stronger correlations with ReQoL and PROMIS – 29 + 2LSS compared with TD versions. For the EQ-5D-5L, the differences in correlations between the TD and LW versions were negligible.

Table 6. Informativity of the EQ-5D-5L and EQ-HWB-9 systems.

Dimensions	H'		J'		% change from TD to LW	
	TD	LW	TD	LW	H' (%)	J' (%)
EQ-5D-5L						
Mobility	0.51	0.53	0.22	0.23	3.9	4.5
Self-care	0.35	0.36	0.15	0.15	2.9	0.0
Usual activities	0.49	0.48	0.21	0.21	–2.0	0.0
Pain/discomfort	0.95	0.97	0.41	0.42	2.1	2.4
Anxiety/depression	1.05	1.07	0.46	0.46	1.9	0.0
EQ-HWB-9						
Getting around inside and outside	0.51	0.57	0.16	0.18	11.8	12.5
Day-to-day activities	0.60	0.63	0.19	0.20	5.0	5.3
Exhausted	1.22	1.23	0.38	0.39	0.8	2.6
Lonely	1.20	1.22	0.38	0.39	1.7	2.6
Concentrating/thinking clearly	1.16	1.18	0.37	0.37	1.7	0.0
Anxious	1.23	1.29	0.39	0.41	4.9	5.1
Depressed	1.17	1.23	0.37	0.39	5.1	5.4
Control	1.19	1.22	0.37	0.39	2.5	5.4
Pain (severity)	0.94	0.98	0.30	0.31	4.3	3.3

H' indicates Shannon index; J', Shannon evenness index; LW, last week; TD, today; *P* value for all correlation coefficients is < .001.

Known-Group Validity

EQ-5D-5L TD and LW LSSs differentiated between 7 and 9 of the 14 known groups, respectively, with moderate or large effect sizes, compared with 8 and 10 groups for the EQ-HWB-9 (Table 4). The largest difference in effect sizes was observed for mental health conditions for LW (EQ-HWB-9: *d* = 1.75 vs EQ-5D-5L: *d* = 1.57).

Structural Validity

PCA results identified 6 components for TD and 5 for LW, explaining 64% and 62% of the total variance, respectively (Table 5). Both TD and LW models identified 5 common factors: (1) mental health, (2) physical health, (3) pain, 4) cognition and quality of life, and (5) ability to participate in social roles and activities. For TD, a distinct “sleep quality” factor emerged from PROMIS – 29 + 2 sleep items, whereas, in LW, sleep items loaded onto the pain factor. The “concentrating/thinking clearly” item of the EQ-HWB-9 did not load onto the cognition and quality of life factor in either model but on the mental health factor. Differences in factor loadings between the TD and LW versions were minimal for both the EQ-5D-5L and EQ-HWB-9.

Informativity

The LW versions of the EQ-5D-5L and EQ-HWB-9 showed higher absolute informativity than TD versions, except for EQ-5D-5L usual activities (Table 6). Regardless of the recall period, 6 of 9 EQ-HWB-9 items had higher absolute informativity than any EQ-5D-5L item. Absolute informativity differences between TD and LW were larger for the EQ-HWB-9 than

EQ-5D-5L, with EQ-HWB-9 “getting around inside and outside” showing the largest difference (10.5%).

Sensitivity Analysis

Given that the sample included disproportionately more participants with higher educational attainment (60% with postsecondary education or above), we conducted a sensitivity analysis by running the psychometric analyses separately for those with secondary education or below and those with postsecondary education or above. No meaningful differences were observed.

Discussion

This study investigated the impact of alternative recall periods on the psychometric properties of the EQ-5D-5L and EQ-HWB-9 in a general population sample in Hong Kong. Overall, the results confirm the validity of the original recall periods for both measures (TD for the EQ-5D-5L and LW for the EQ-HWB-9). Consistent with the literature, more problems were reported using the longer recall period on both measures.^{7,14} The differences in distribution for items with different recall periods were most pronounced for constructs such as pain, mental health, cognition, and loneliness. The psychometric properties of the EQ-5D-5L and EQ-HWB-9 were similar between the LW and TD recall periods. However, the LW versions demonstrated modestly better performance, exhibiting lower ceiling, improved informativity, and stronger convergent and known-group validity compared with the TD versions. Although the impact of recall was overall small, it was larger for the EQ-HWB-9 than for EQ-5D-5L. This may be due to both the different constructs captured by these measures and the different response scales (eg, severity vs frequency). These findings potentially suggest that a longer recall period may more effectively capture the variability in health and well-being. A week-long recall allows respondents to reflect on a broader range of experiences, potentially smoothing out transient fluctuations in health or well-being that a single day focus might exaggerate or miss.³

The difference in sensitivity of certain health and well-being domains to recall periods might be attributed to their inherent temporal variability and the way individuals process and report experiences over time.³⁷ Affective and psychosocial constructs, which are often dynamic and influenced by fleeting contextual factors, tend to benefit from longer recall periods because these allow respondents to average out episodic fluctuations, mitigate recency bias, and capture a more representative spectrum of experiences.⁷ In contrast, physical domains that reflect more stable, often chronic states are less affected by recall duration, aligning well with shorter TD assessments that provide a reliable snapshot without significant distortion from daily variability.³⁸ These insights emphasize the need to align recall periods with the measured constructs.³⁹ For example, loneliness often emerges as an episodic emotional state shaped by transient social interactions, which may not manifest consistently on a single day but build up over a week, potentially leading to higher reported frequencies in longer recalls.⁴⁰ Similarly, exhaustion fluctuates with short-term stressors, sleep quality, or energy levels, risking under- or overestimation in a TD frame if the day is unusually calm or demanding, whereas a weekly period may yield a balanced view of these variations. Usual activities, however, typically represent enduring physical capabilities that remain steady day-to-day for many people, thus showing minimal differences across recall lengths and indicating its stability relative to more variable psychosocial items.

A limitation of this study is the use of a relatively healthy general population sample, which may restrict the generalizability of our findings. Our sample exhibited low levels of health and well-being impairment, with very few participants reporting issues on the first 3 EQ-5D-5L dimensions or certain EQ-HWB-9 items. As a result, the psychometric performance observed here may not be generalizable to more diverse or clinically burdened groups, for which the instrument is also intended to be applied. Although the EQ-HWB-9 instrument was primarily designed for use among patients, social care users, and carers, which are the populations that typically experience greater health and well-being challenges than the general public, our approach of recruiting from a general population provides valuable initial insights into the instrument's performance in a broader, less impaired context, which can inform future research in this area. Moreover, the study design might have also influenced the results because respondents completed a total of 2 EQ-5D-5L and 2 EQ-HWB-9 with different recall periods each. Despite successful randomization, responses on the same instrument with a different recall period may have been influenced by previous ones. Last, although the exclusion criteria improved the data quality, they might have biased the sample toward younger, digitally literate participants, which could further affect the generalizability of our findings.

Conclusions

The observed differences in psychometric properties between the TD and LW recall periods were generally small in this general population sample from Hong Kong. Overall, the results provide evidence supporting the validity of the original recall periods for both the EQ-5D-5L and EQ-HWB-9. Further research is needed in the intended areas of use for each instrument (eg, patients, carers, or social care users) and across different countries to comprehensively assess the merits of the alternative recall period versions of the EQ-5D-5L and EQ-HWB-9 to make informed recommendations to decision makers regarding the contexts in which deviations from the original recall periods may be acceptable.

Author Disclosures

Author disclosure forms can be accessed below in the [Supplemental Material](#) section.

Supplemental Material

Supplementary data associated with this article can be found in the online version at <https://doi.org/10.1016/j.jval.2026.01.001>.

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